THE UNIVERSAL PROPERTY OF THE CANONICAL MAP TO THE SHEAFIFICATION

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file sites-cohomology.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The proof is the author's complete argument; every step the author writes out is inside the counted span and no closing sentence is a placeholder.

Editorial note. Every labelled reference occurring inside the counted spans is carried in assembly.external_references as a non-load-bearing navigation pointer (same-chapter ones at their definition site, cross-chapter ones at the invocation inside the counted span), so no load-bearing mathematics is replaced by a stub. No bibliography citation occurs inside the counted spans.

Editorial note. The checked chapter list for same-name collisions is the set of stacks chapters staged in this repository: chow, cohomology, divisors, etale-cohomology, flat, more-algebra, more-morphisms, morphisms, spaces-more-morphisms, varieties together with this card's chapter sites-cohomology.

Editorial note. The preamble is the chapter's own retained minimal preamble (sources/stacks-sites-cohomology/retained-preamble.tex), a verbatim excerpt of the native preamble: the Xy-pic setup and the native theorem declarations with the mathematical macros.

Allowed standard prerequisites (editorial).

- Functoriality of the plus construction. Same chapter, Lemma lemma-plus-functorial. Produces the commutative diagram comparing the plus constructions of the presheaf and of the sheaf.
- The plus construction on a sheaf is an isomorphism. Same chapter, Theorem theorem-plus. Makes the lower horizontal maps of the diagram isomorphisms.
- Sections of the sheafification are locally sections of the presheaf. Same chapter, Lemma lemma-plus-surjective. Gives uniqueness of the factorization.

Source reference. Original lemma 00W5: sources/stacks-sites/pinned-original-source.tex, source lines 2279–2279. Navigation pointer invoked at proof line 2279; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Source reference. Original theorem 00WB: sources/stacks-sites/pinned-original-source.tex, source lines 2295–2295. Navigation pointer invoked at proof line 2295; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Date: stacks-project:sites.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Source reference. Original lemma 00W9: sources/stacks-sites/pinned-original-source.tex, source lines 2218–2218. Navigation pointer invoked at proof line 2218; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Proposition 0.1. *The canonical map $\mathcal{F} \to \mathcal{F}^\#$ has the following universal property: For any map $\mathcal{F} \to \mathcal{G}$, where $\mathcal{G}$ is a sheaf of sets, there is a unique map $\mathcal{F}^\# \to \mathcal{G}$ such that $\mathcal{F} \to \mathcal{F}^\# \to \mathcal{G}$ equals the given map.*

Proof. By Lemma 00W5 we get a commutative diagram

$$\begin{array}{ccccc} \mathcal{F} & \longrightarrow & \mathcal{F}^+ & \longrightarrow & \mathcal{F}^{++} \\ \downarrow & & \downarrow & & \downarrow \\ \mathcal{G} & \longrightarrow & \mathcal{G}^+ & \longrightarrow & \mathcal{G}^{++} \end{array}$$

and by Theorem 00WB the lower horizontal maps are isomorphisms. The uniqueness follows from Lemma 00W9 which says that every section of $\mathcal{F}^\#$ locally comes from sections of $\mathcal{F}$. $\square$